PASSWORD STRENGTH CHECKER USING PYTHON (Report)

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Project Title
Metadata Extractor using Kali Linux
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Introduction

Passwords are the first line of defense for protecting user accounts and sensitive information. Weak passwords can be easily guessed or cracked by attackers, leading to unauthorized access. A Password Strength Checker is a tool that evaluates the strength of a password based on factors such as length, uppercase letters, lowercase letters, numbers, and special characters. This project is developed using Python on Kali Linux to help users create secure passwords.

Objective

- To develop a password strength checking system.
- To analyze passwords based on security criteria.
- To classify passwords as Weak, Medium, or Strong.
- To improve awareness of password security.

Software Requirements

- Kali Linux
- Python 3
- Nano Editor
- Terminal

Algorithm

1. Start the program.
2. Accept a password from the user.
3. Check the password length.
4. Check for uppercase letters.
5. Check for lowercase letters.
6. Check for numbers.
7. Check for special characters.
8. Calculate the password score.
9. Display the password strength.

10. End the program.

Source Code

```
import re
password = input("Enter Password: ")
score = 0
if len(password) >= 8:
score += 1
if re.search("[A-Z]", password):
score += 1
if re.search("[a-z]", password):
score += 1
if re.search("[0-9]", password):
score += 1
if re.search("[!@#$%^&*]", password):
score += 1
if score <= 2:
print("Weak Password")
elif score <= 4:
print("Medium Password")
else:
print("Strong Password")
```

Execution Steps

1. Open Terminal in Kali Linux.
2. Create a project folder.
3. Create the Python file named password_checker.py.
4. Write the Python code using Nano Editor.
5. Save the file.
6. Execute the program using: python3 password_checker.py
7. Enter a password.
8. View the result.

Output

Example 1
Enter Password: abc
Output: Weak Password
Example 2
Enter Password: abc12345
Output: Medium Password
Example 3
Enter Password: Admin@123
Output: Strong Password

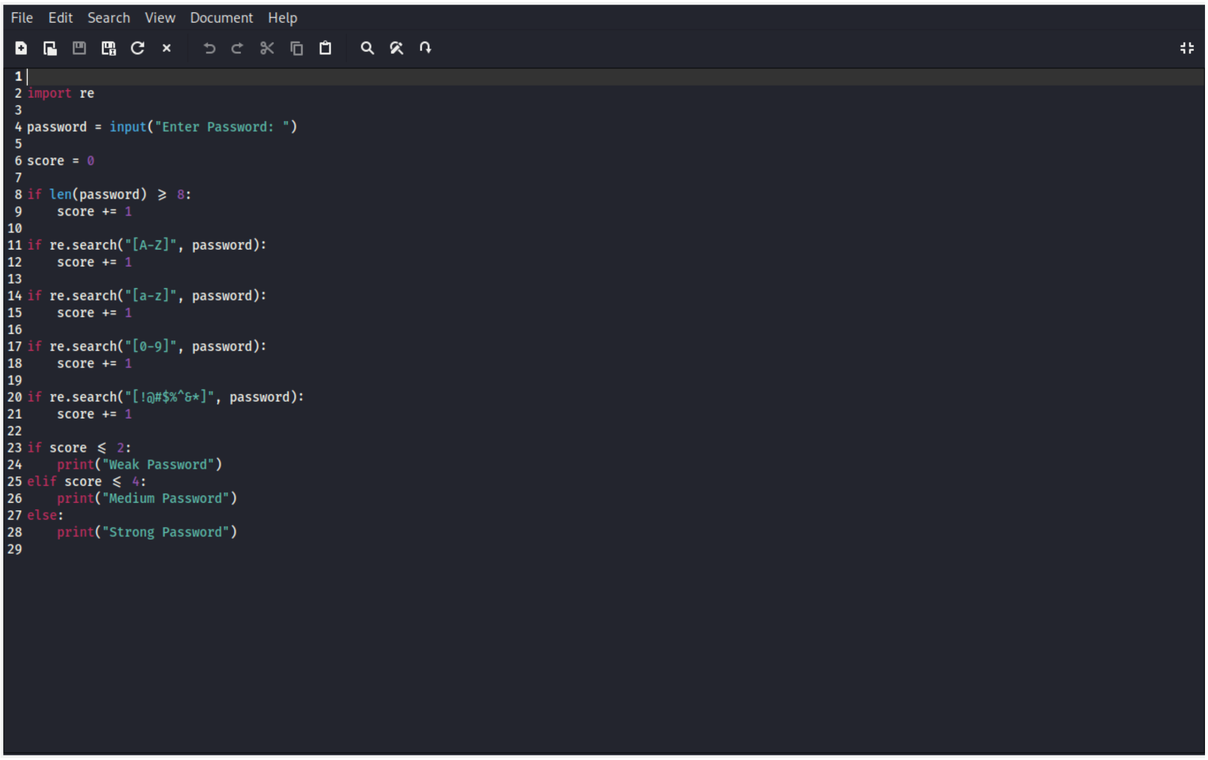
Advantages

- Easy to use.
- Improves password security.
- Helps users create strong passwords.
- Reduces the risk of unauthorized access.

Conclusion

The Password Strength Checker successfully evaluates password strength and
classifies
passwords as Weak, Medium, or Strong. The project demonstrates the use of Python
programming
and basic cybersecurity concepts on Kali Linux. It helps users understand the
importance of
creating secure passwords and protecting their accounts from security threats.

Screen Shot Of Code




```python

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Output Screenshot